VeriSLO IP Final Presentation

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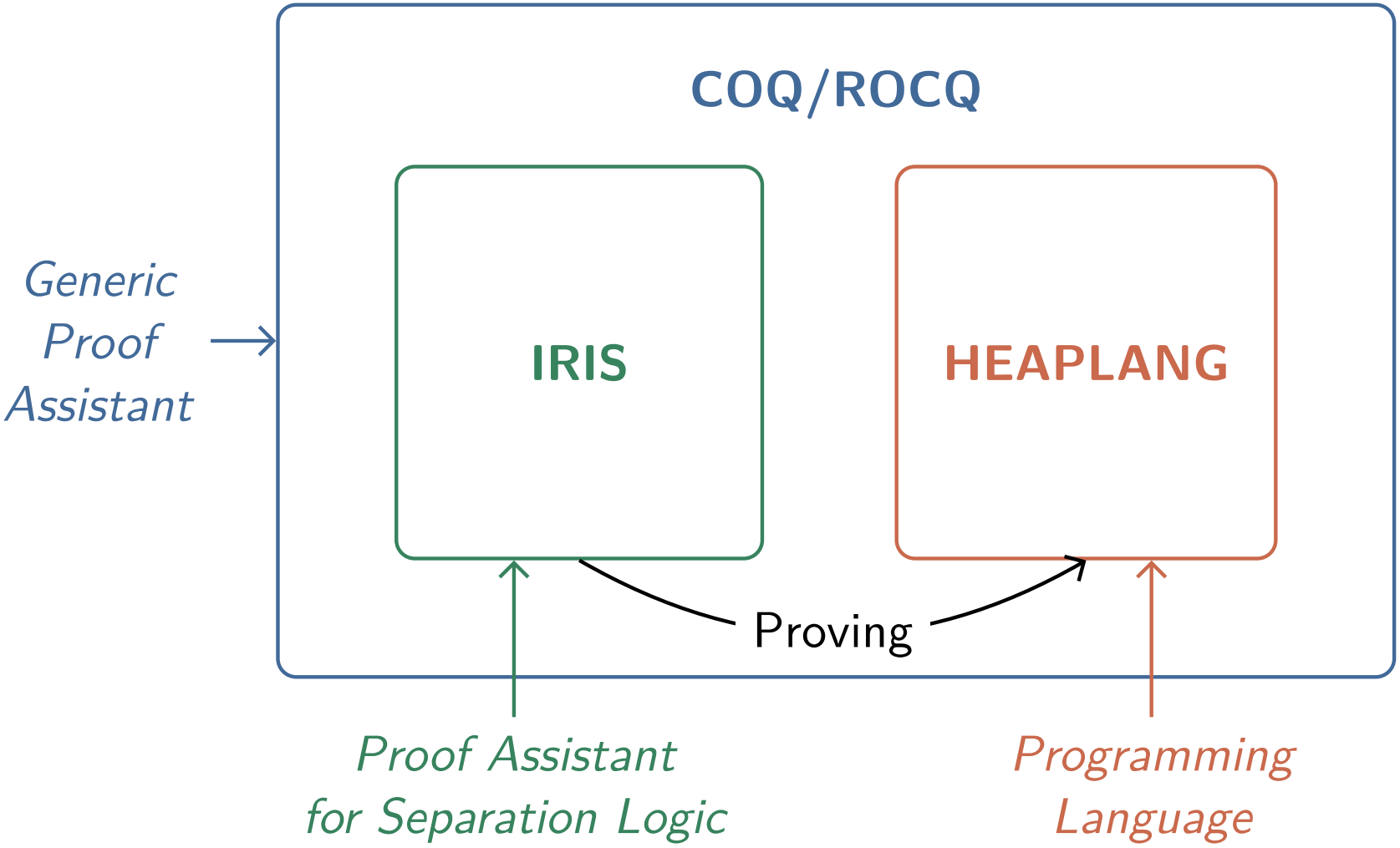
École Normale Supérieure de Lyon, IP Project

March, 17th 2025

Rocq and Iris



The “normal” process

Some OCaml program

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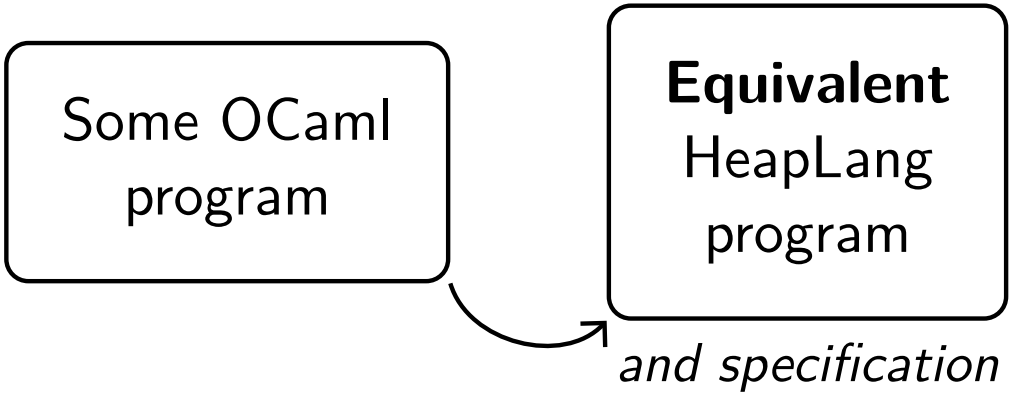

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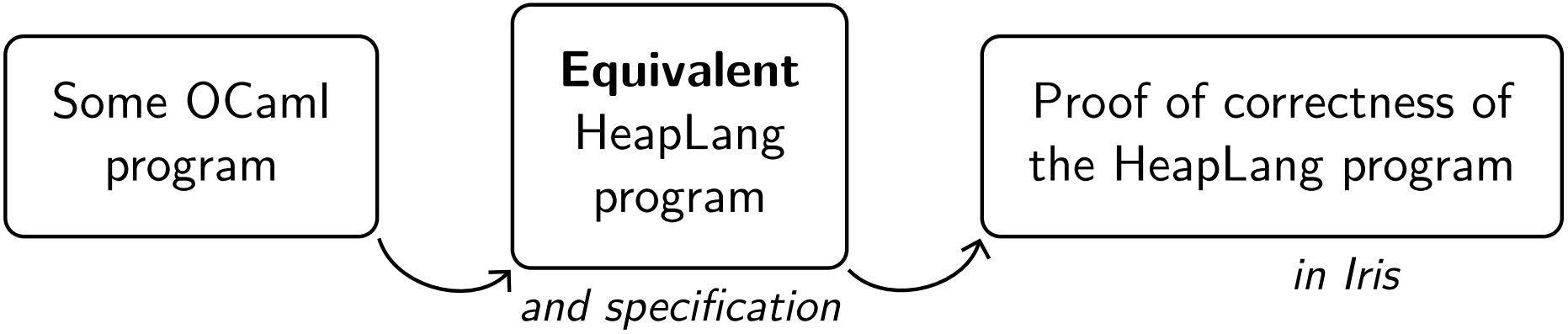

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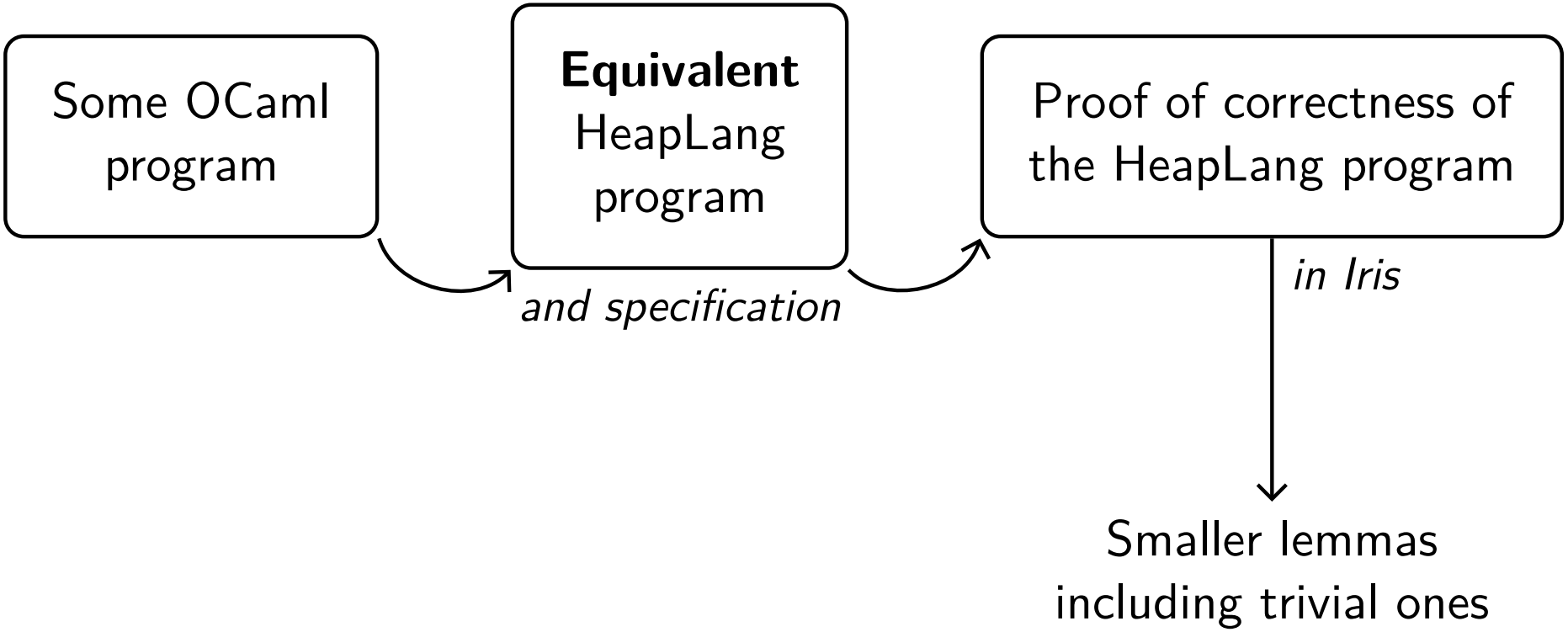

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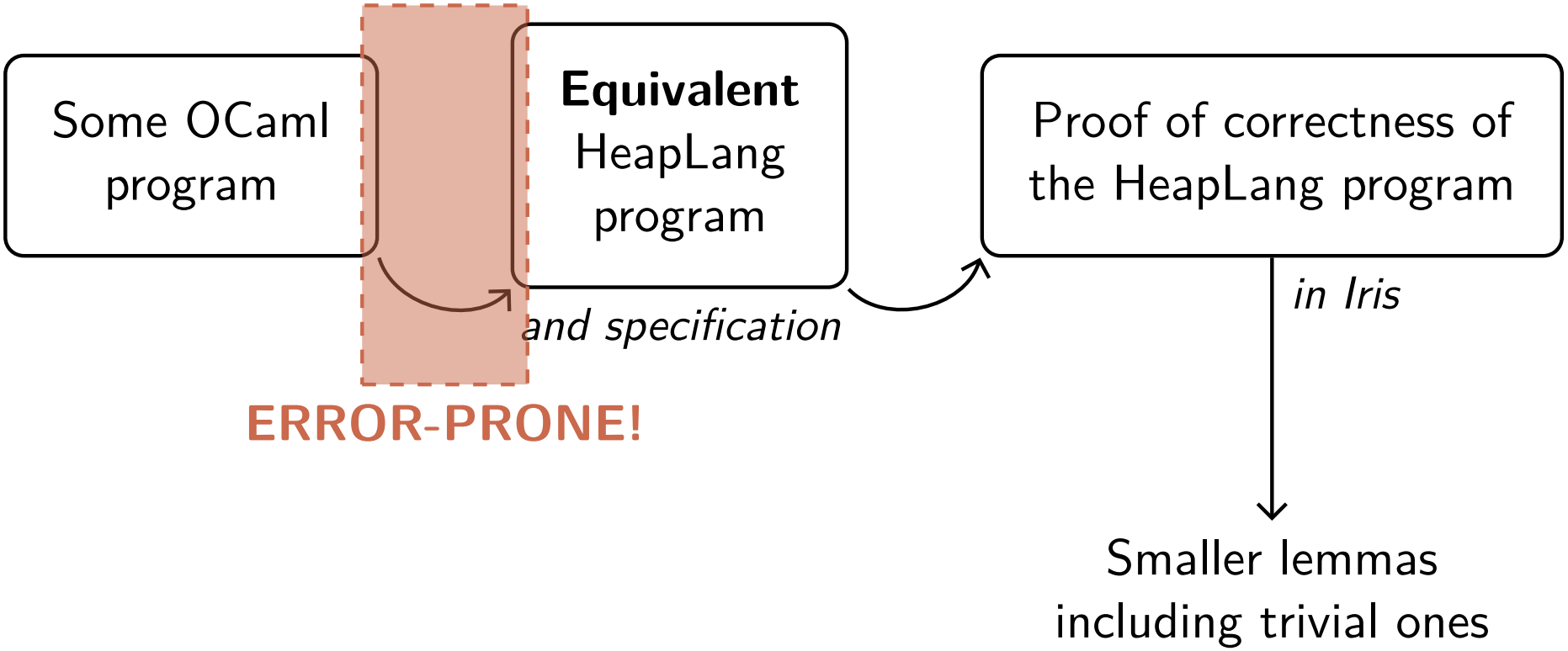

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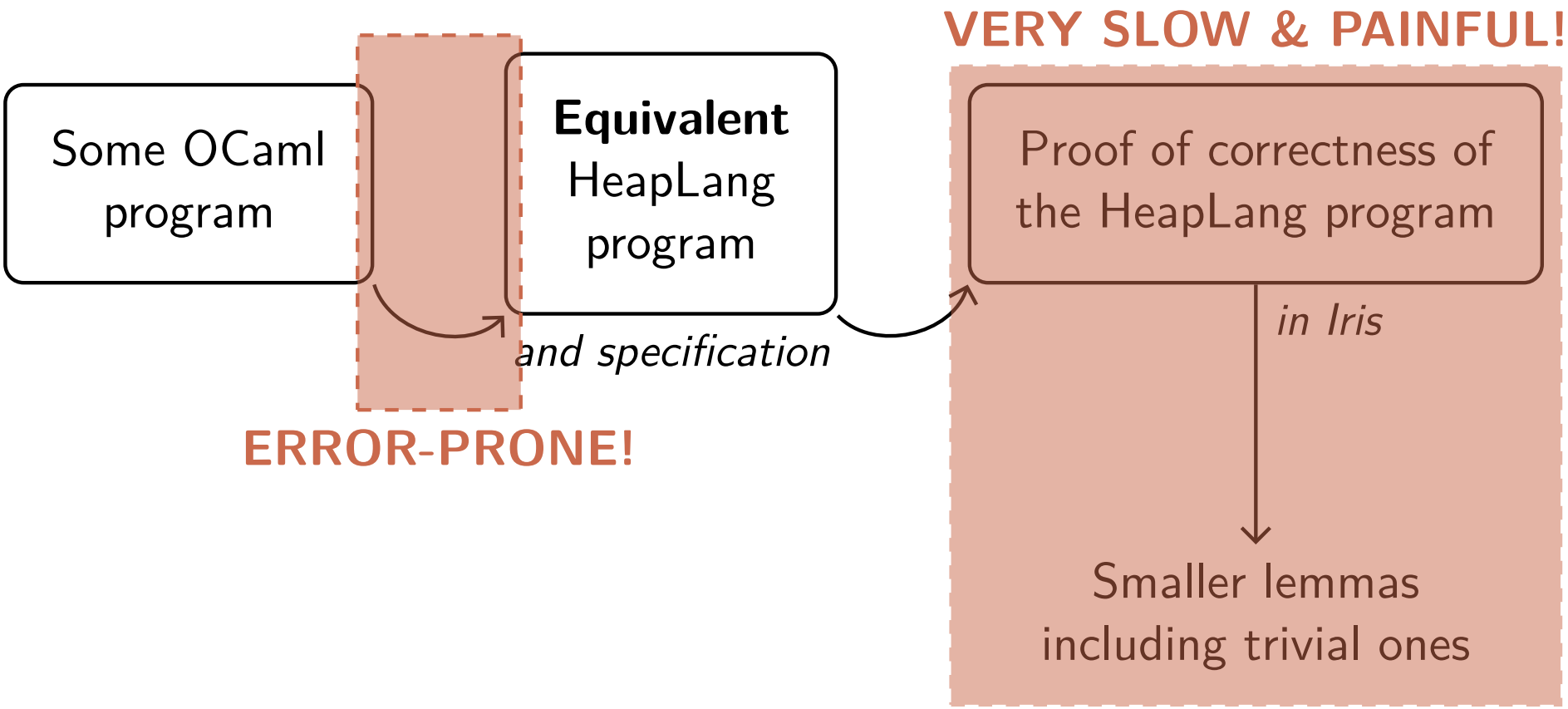

Outline

1. What is VeriSLO?
2. How It Works?
3. New features!
4. Live demonstration

Two example programs

We are able to *automatically* prove programs.

```
let result =
    let x = ref 1 in
    if !x <= 2
        then x := 3;
    !x
    [@post "!x = #3"]
```

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```
let result =
    let x = ref 1 in
    if !x <= 2
        then x := 3;
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    [@post "!x = #3"]
```

```
[@@@vernac {|
Definition even (n : Z) : Prop :=
  exists k : Z, n = (2 * k)%Z.
|}]
```

```
let result =
  let x = ref 0 in
  while !x <= 10 do
    x := !x + 2;
  done
  [@invariant "even !x"]
  [@post "!x = #12"]
```

Two example programs

We are able to *automatically* prove programs.

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    let x = ref 1 in
    if !x <= 2
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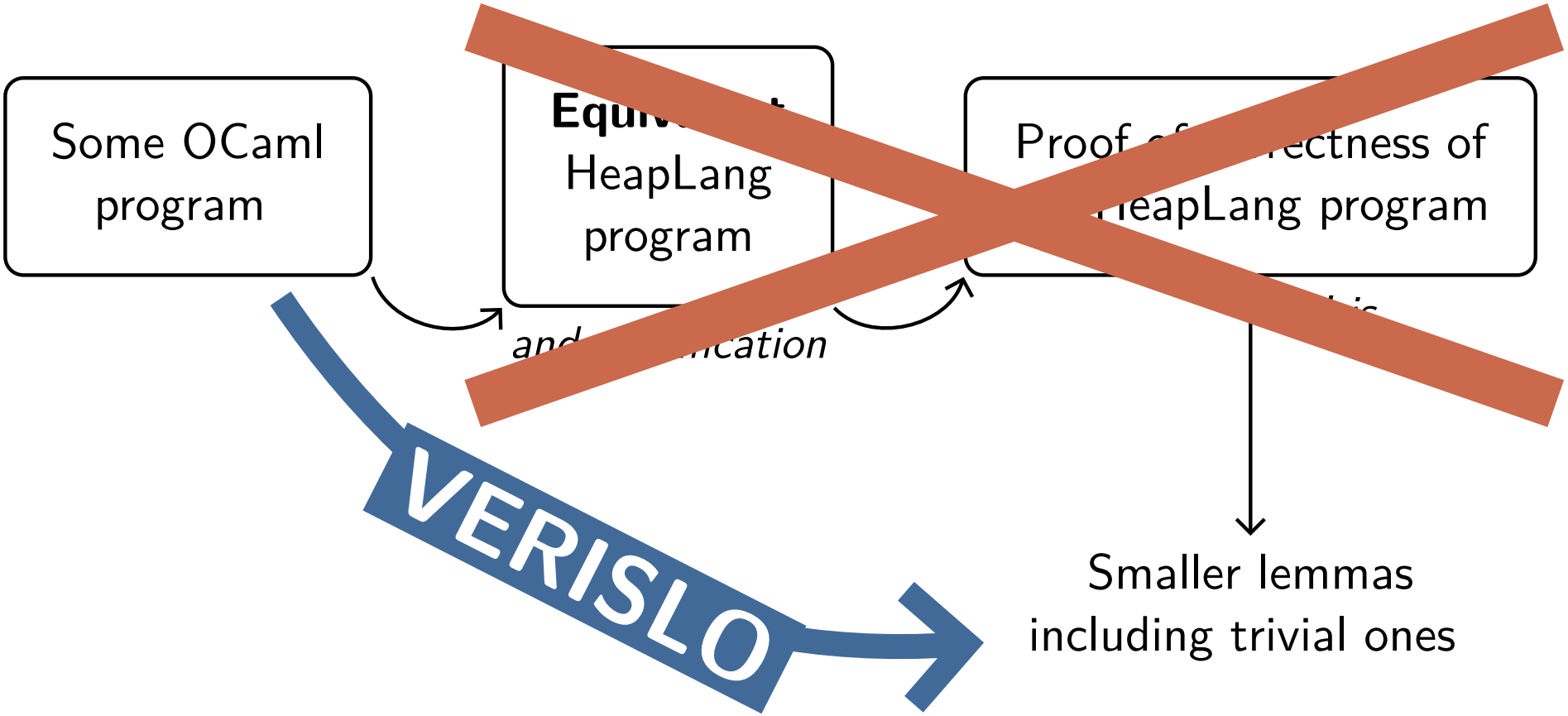
let result =
  let x = ref 0 in
  while !x <= 10 do
    x := !x + 2;
  done
  [@invariant "even !x"]
  [@post "!x = #12"]
```

We have some ***syntactic sugar***:

$$\texttt{even !x} \quad \rightsquigarrow \quad \exists v : \texttt{val}, \quad \texttt{x} \mapsto v \quad * \quad \texttt{even}\ v.$$

The VeriSLO process



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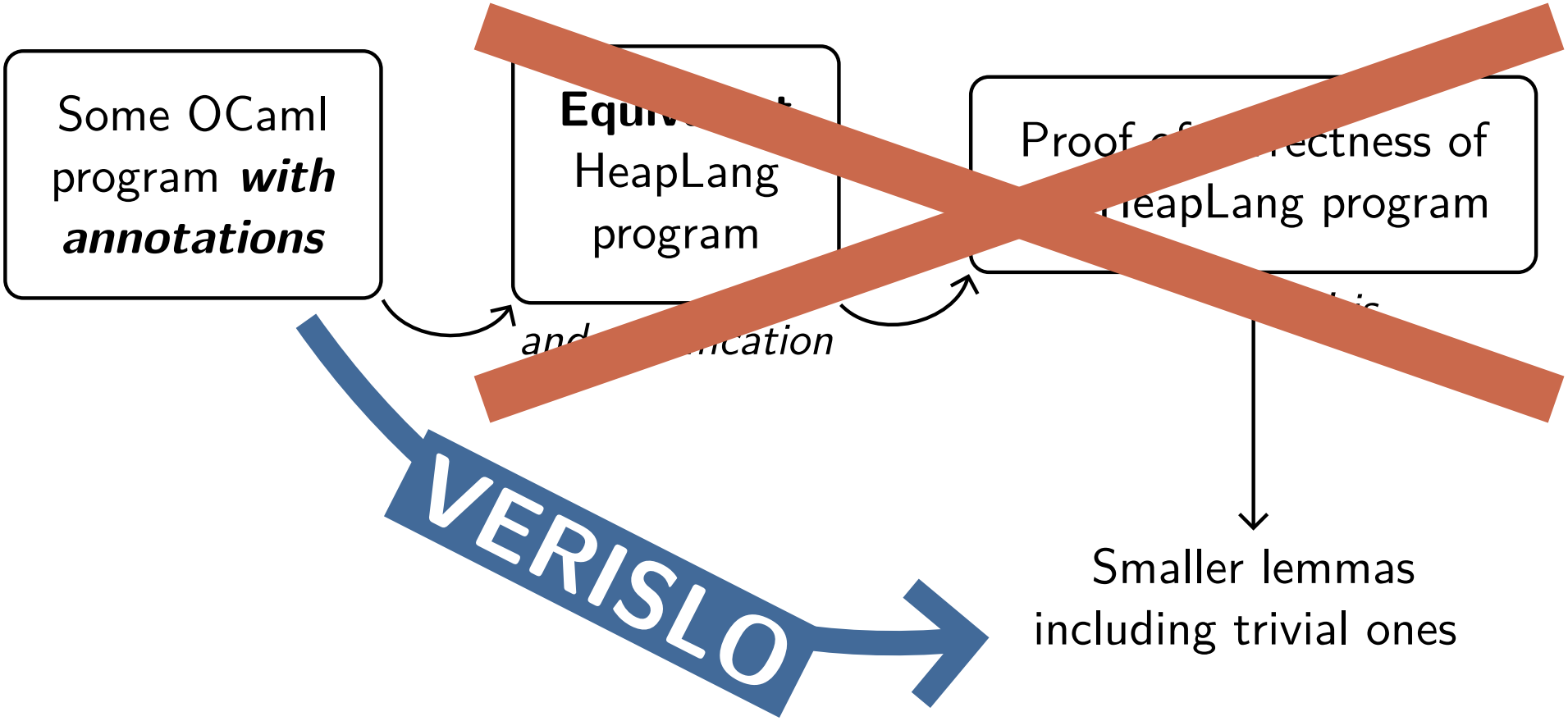

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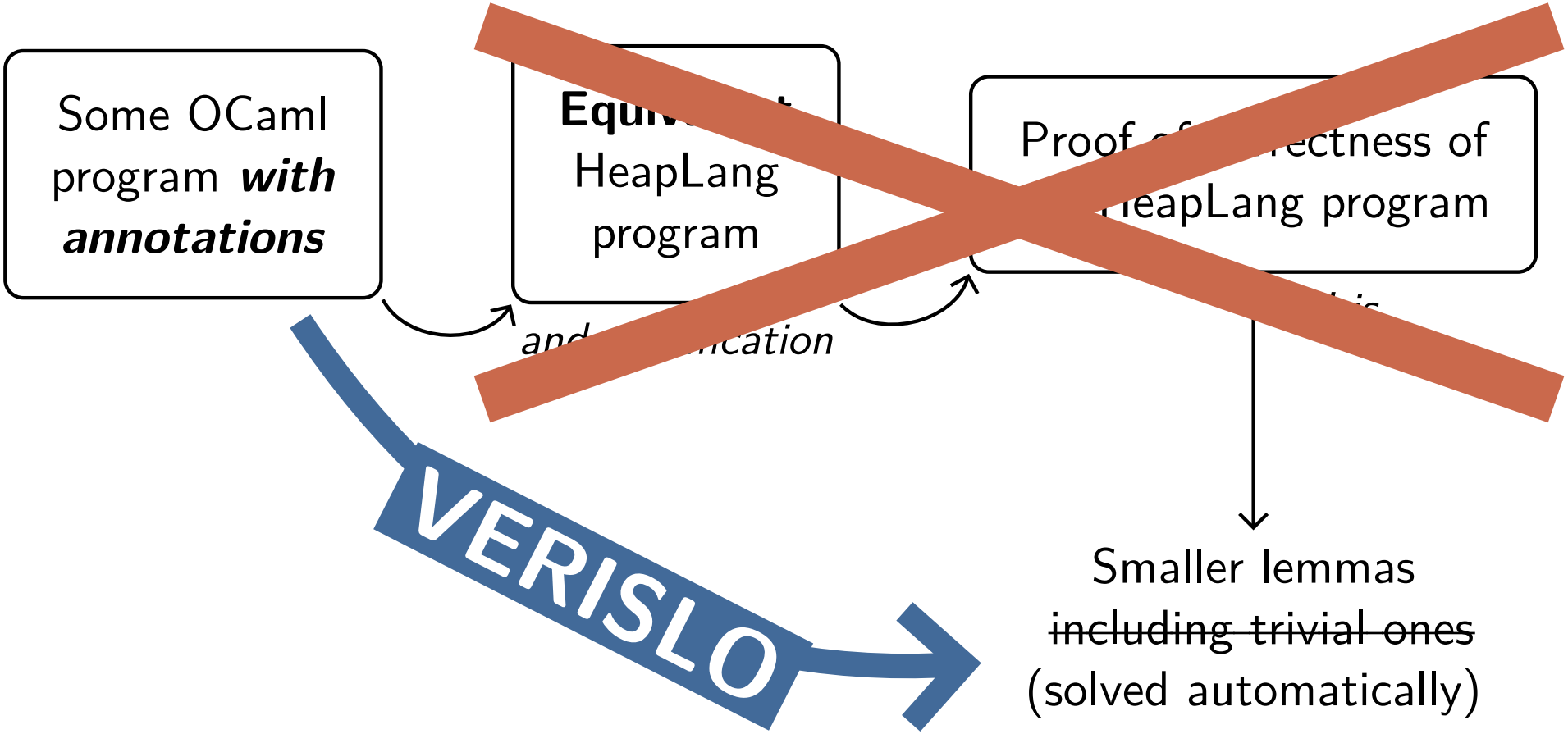

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Architecture of VeriSLO: A bird's eye view

Annotated OCaml program

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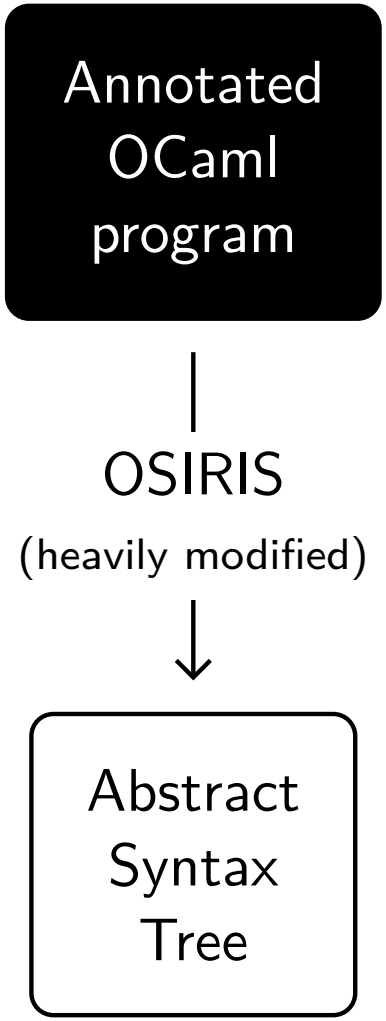

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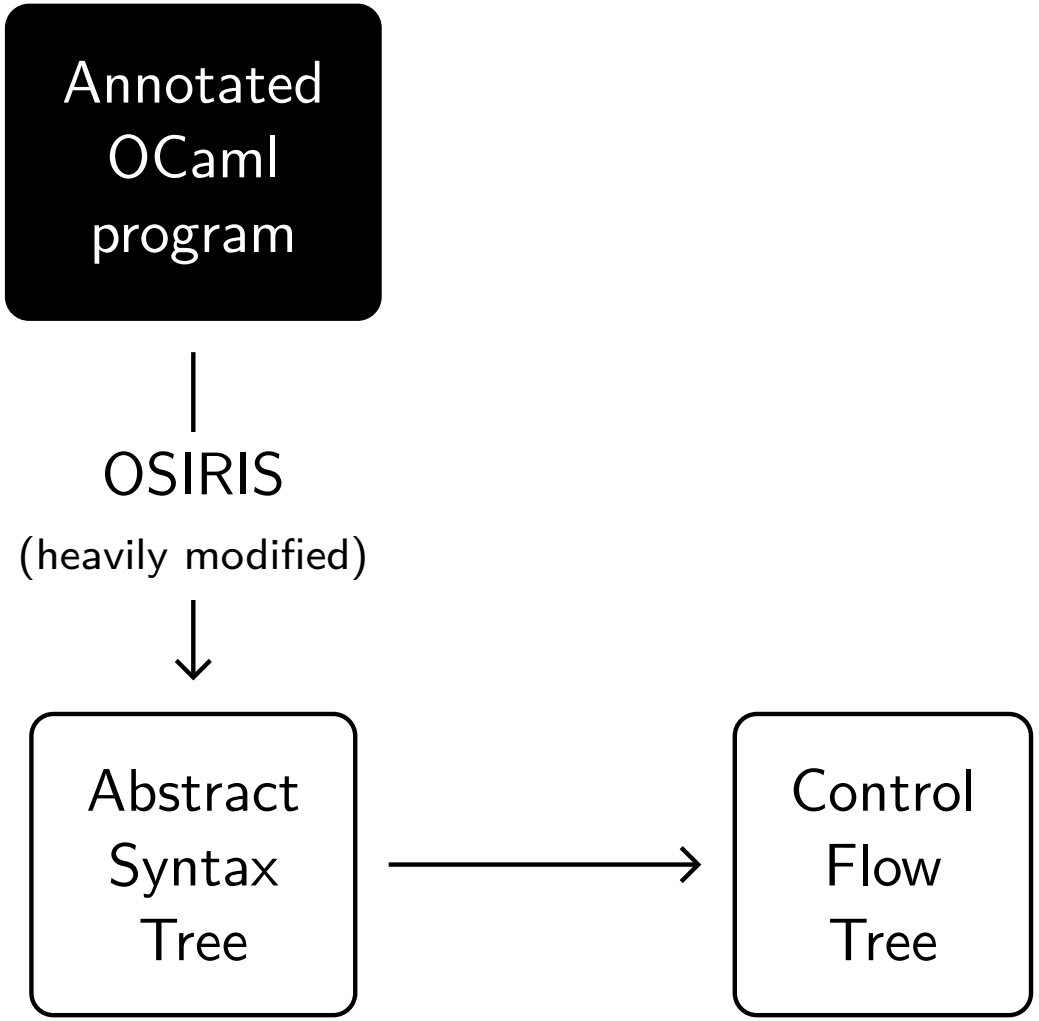

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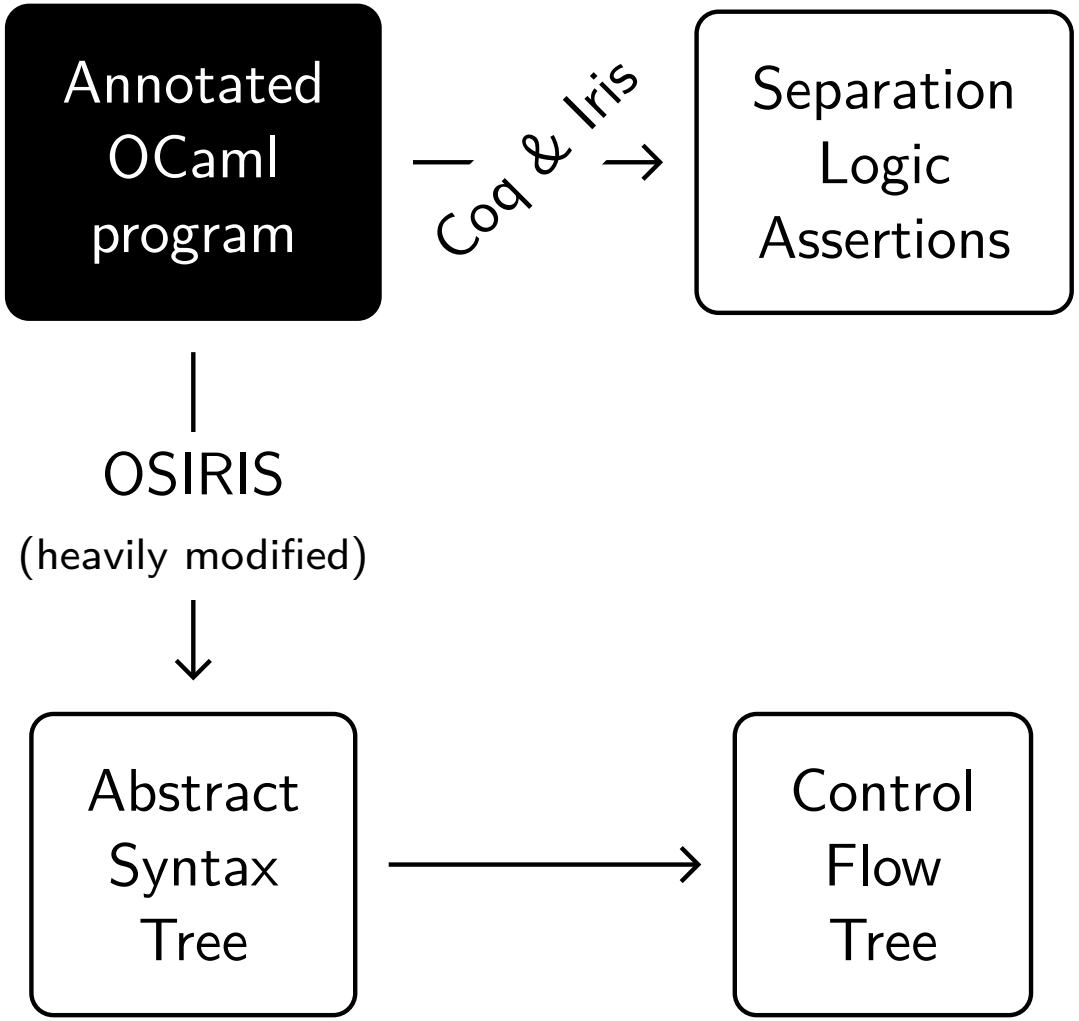

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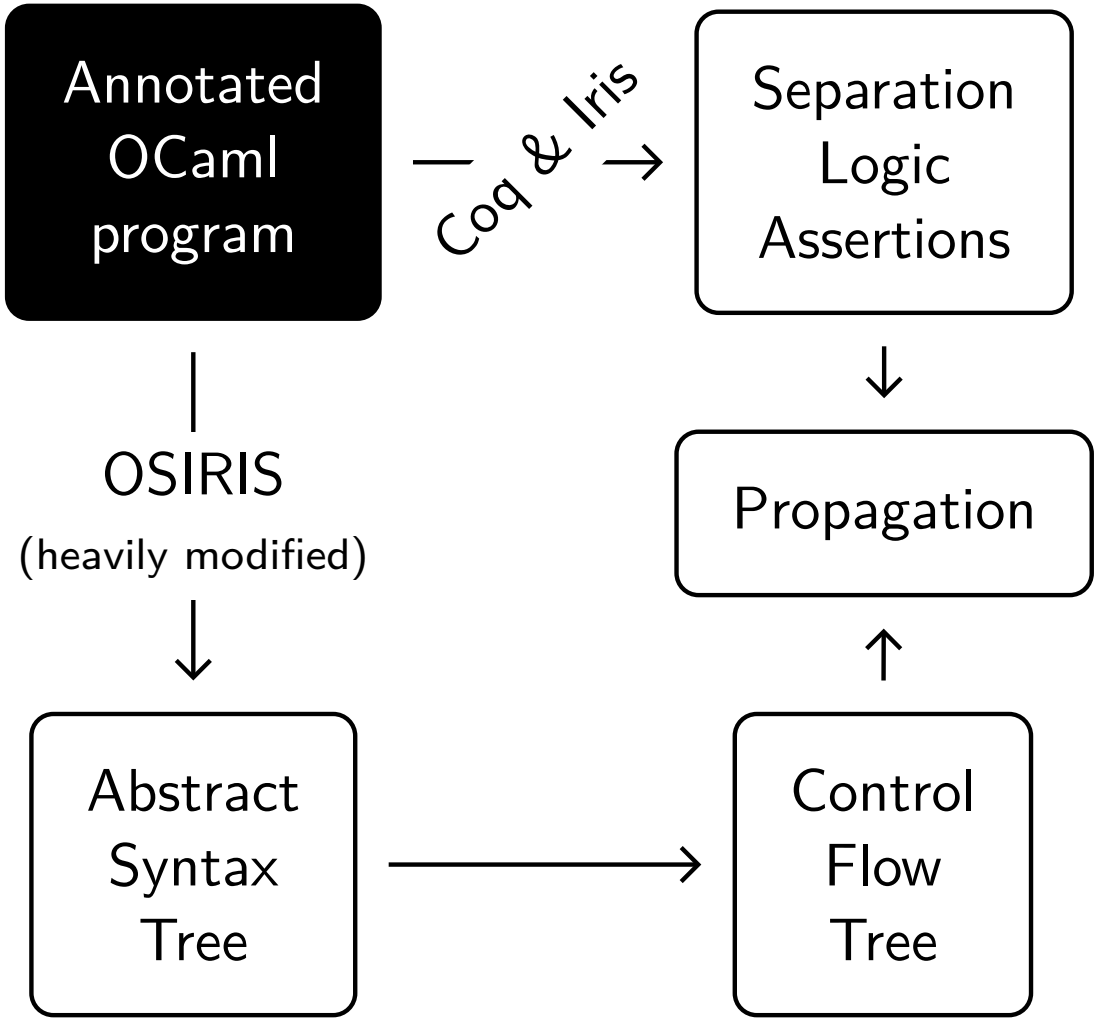

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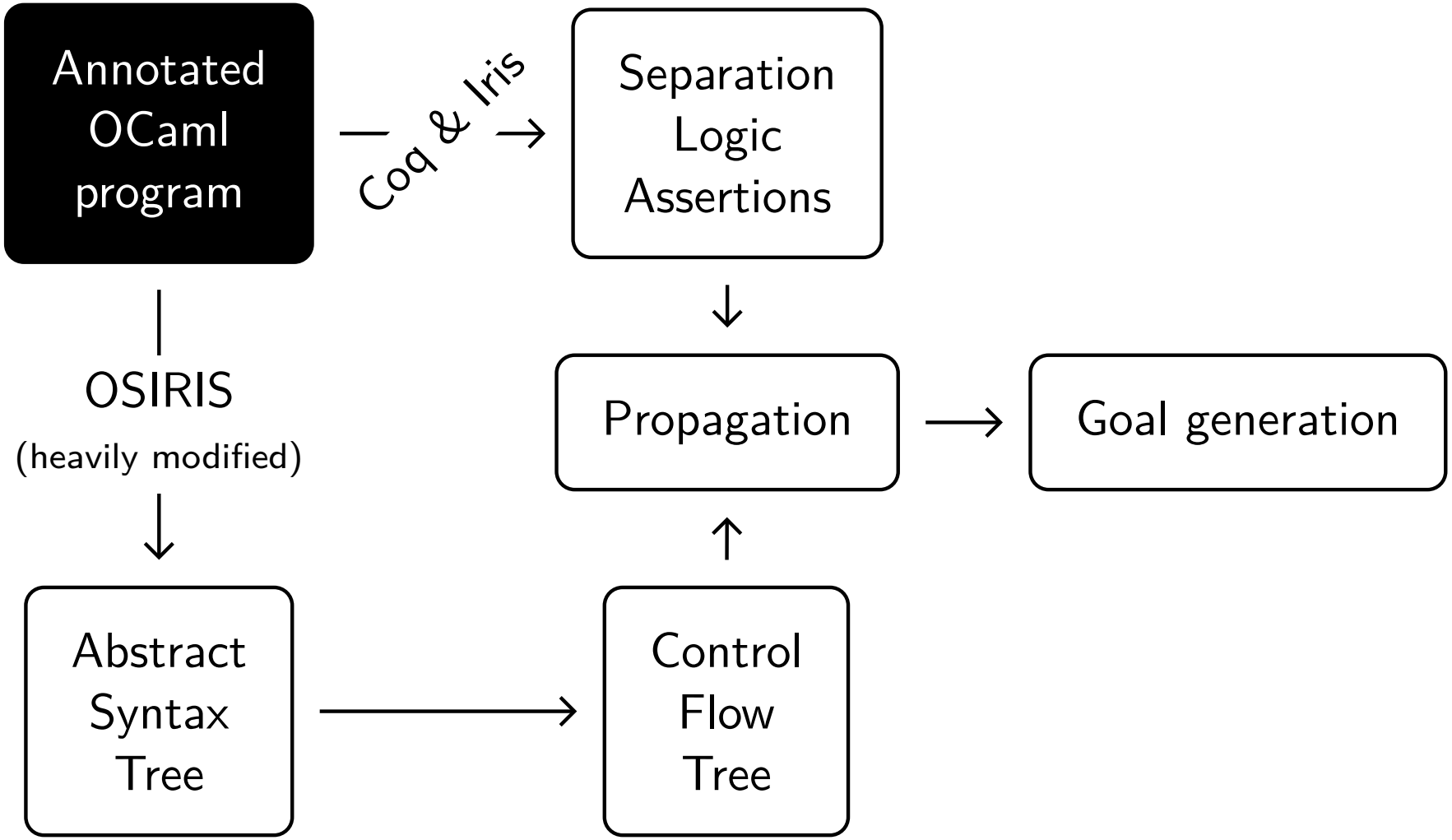

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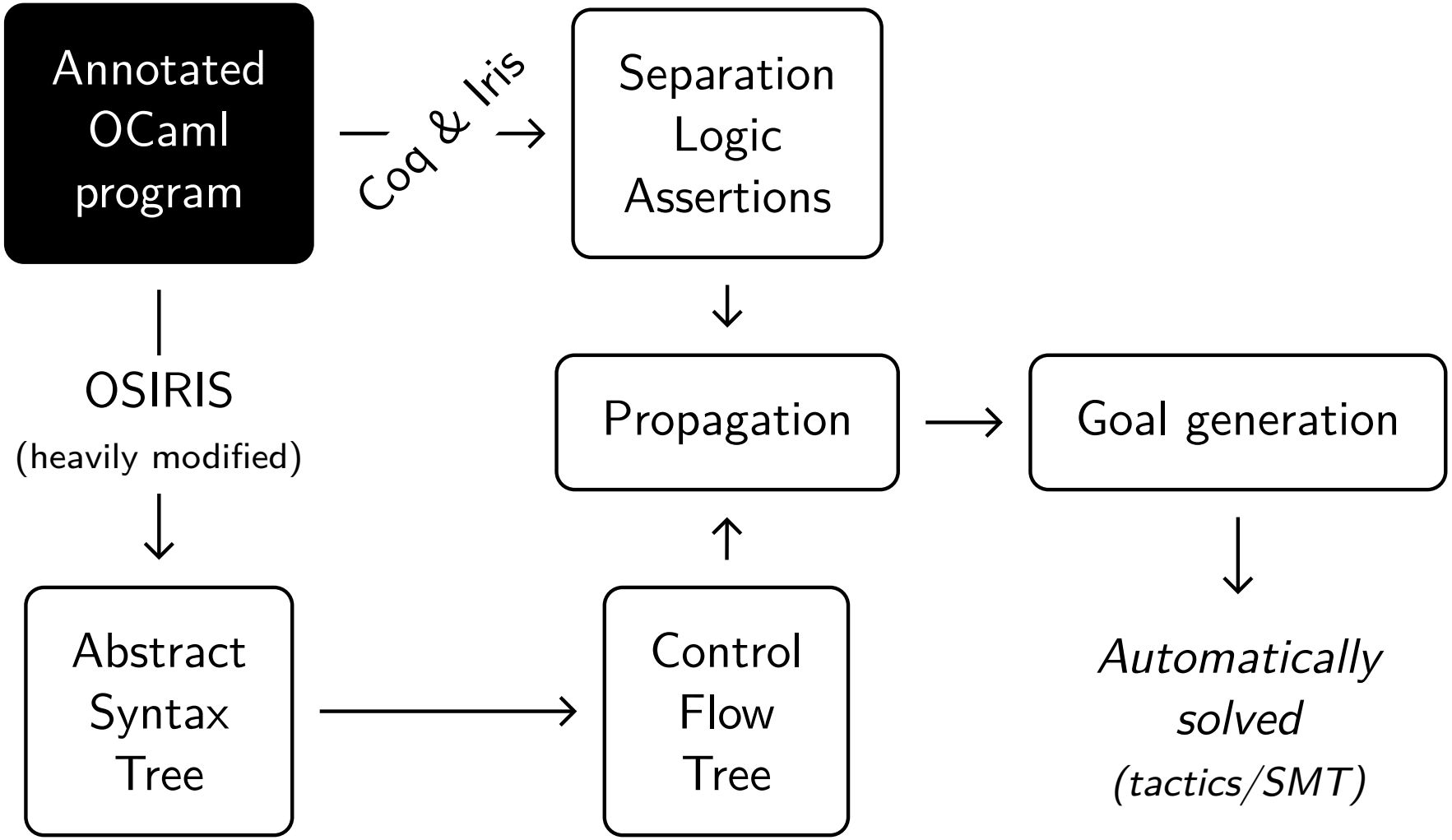

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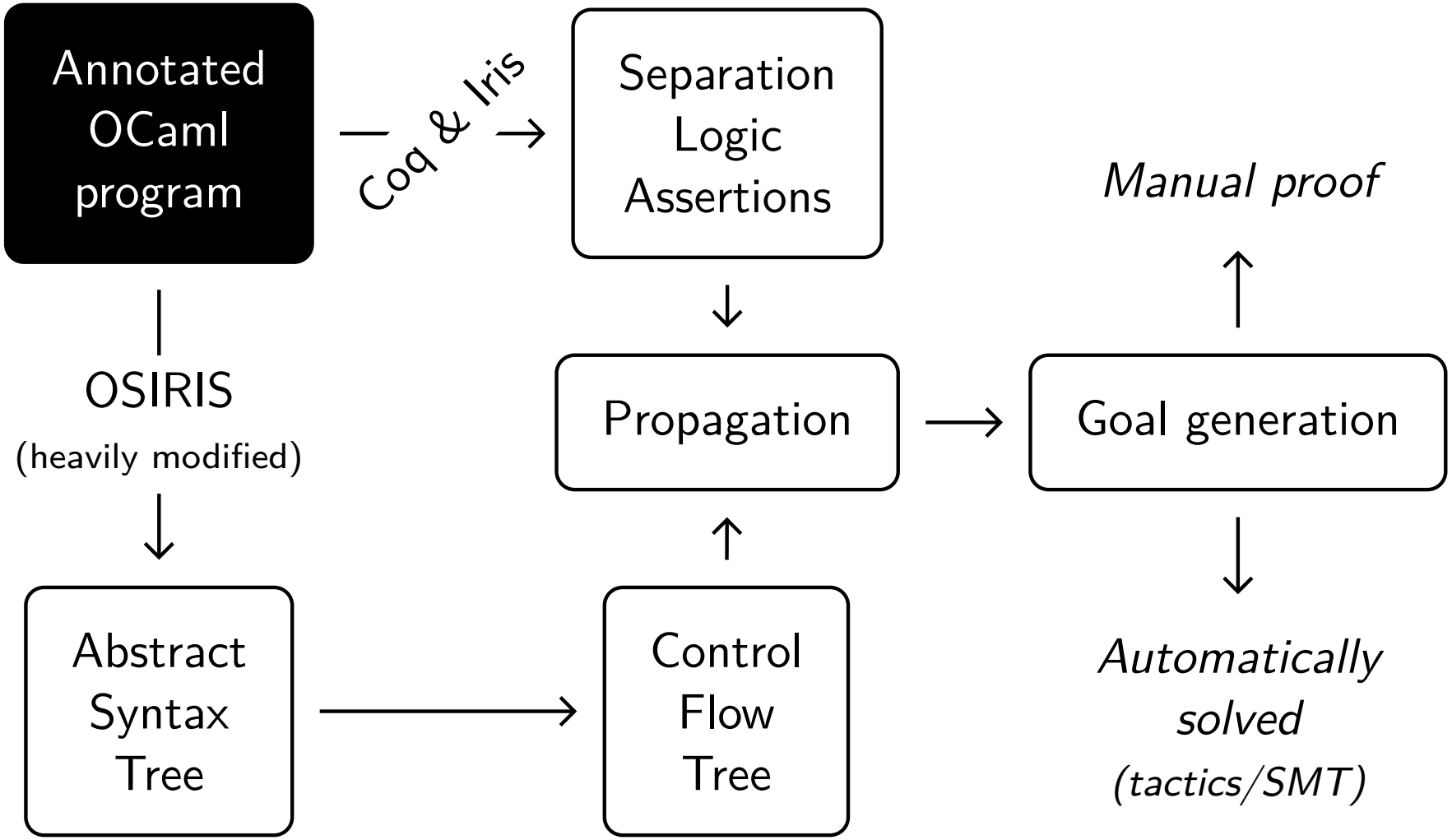

Control Flow Tree (CFT)

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let result =
    let x = ref 1 in
    if !x <= 2
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    !x
    [@ret "y"] [@post "y = #3"]
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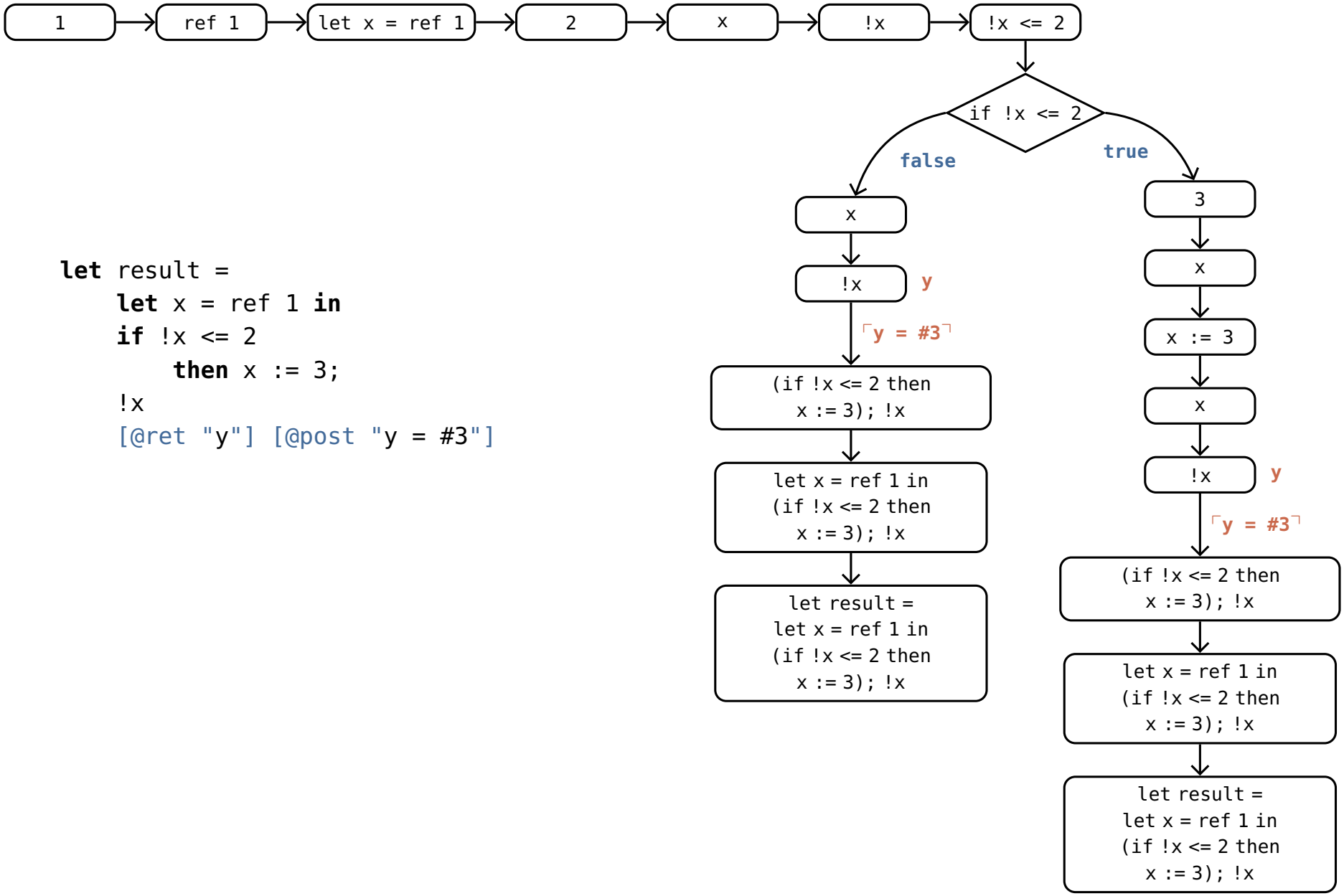

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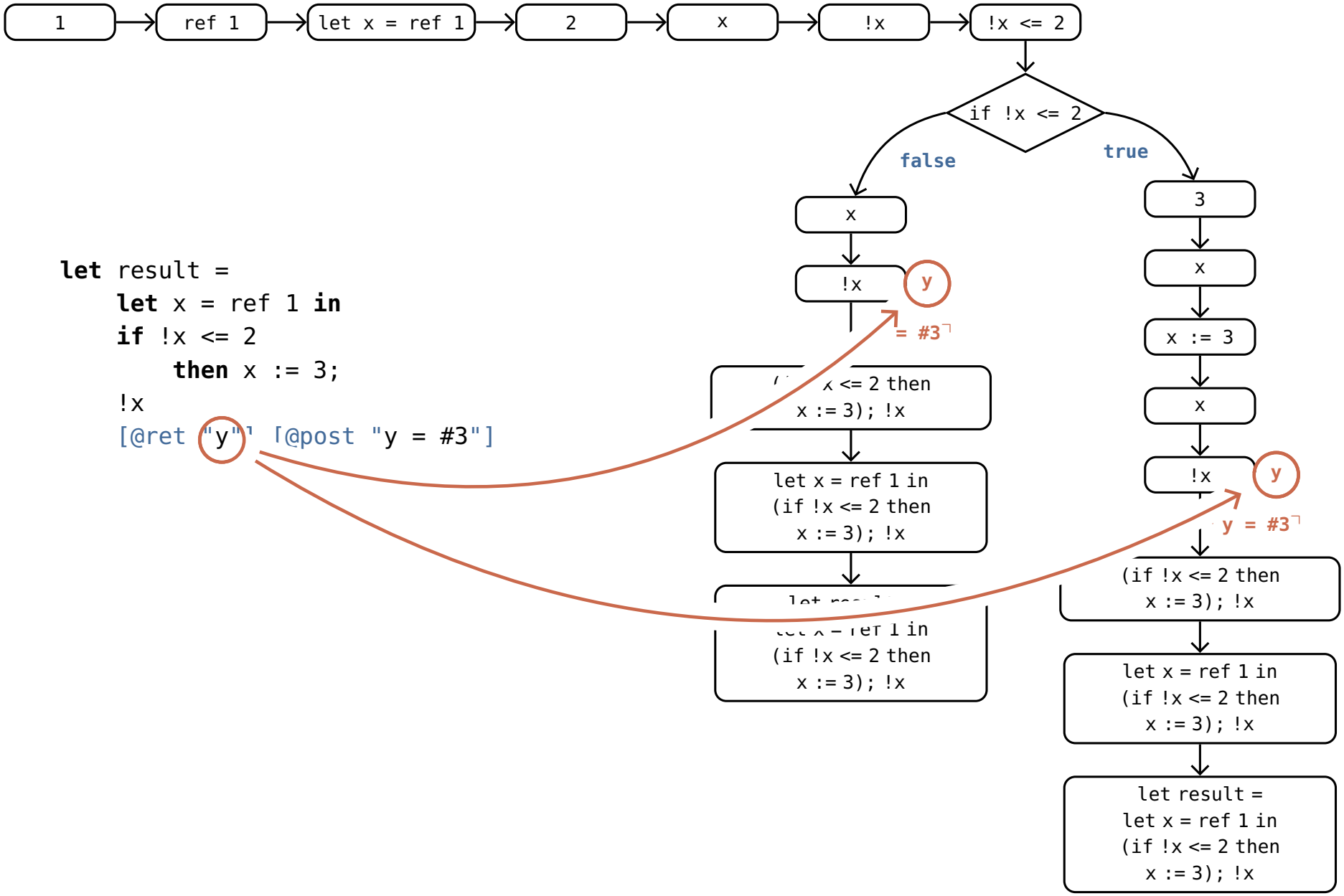

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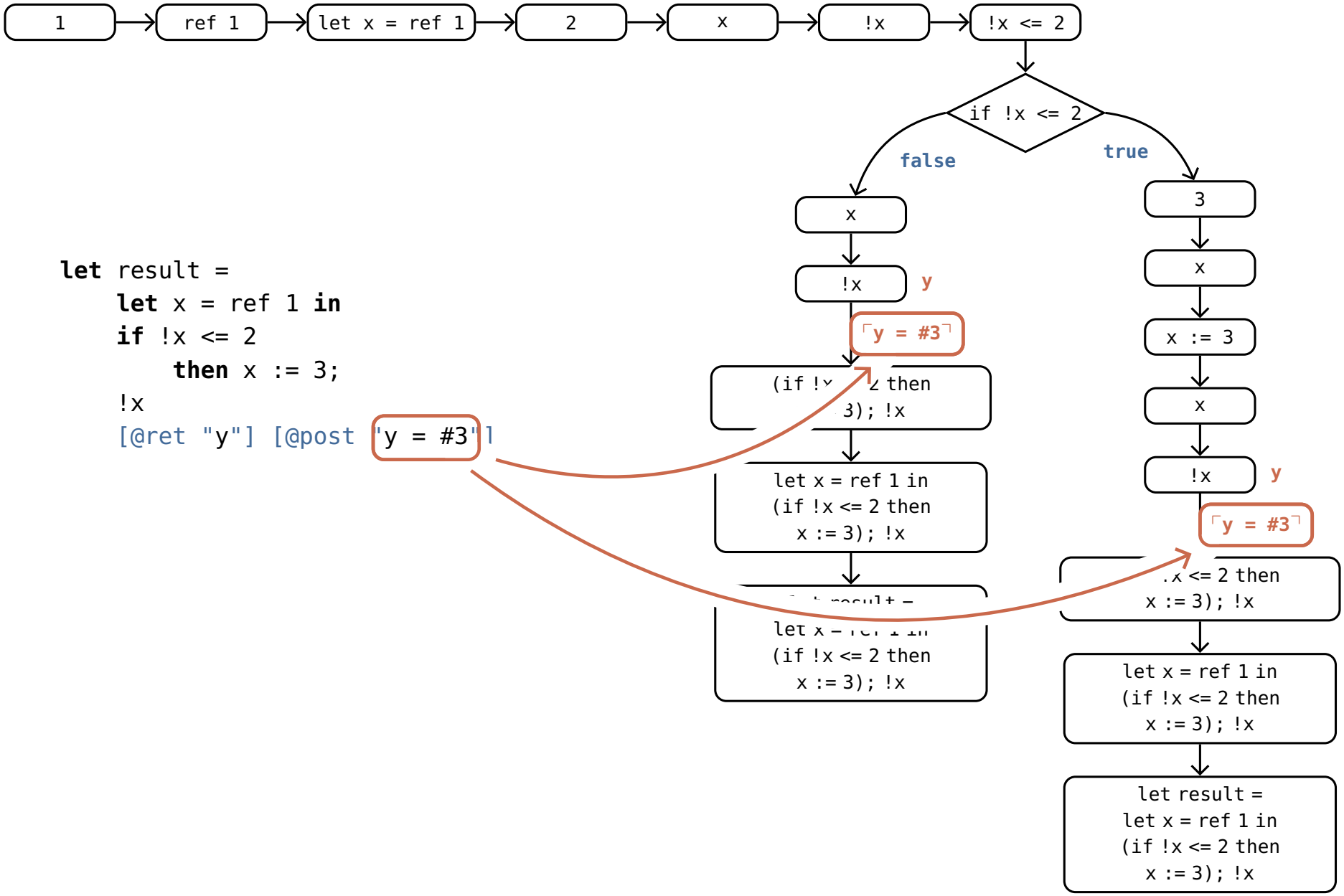
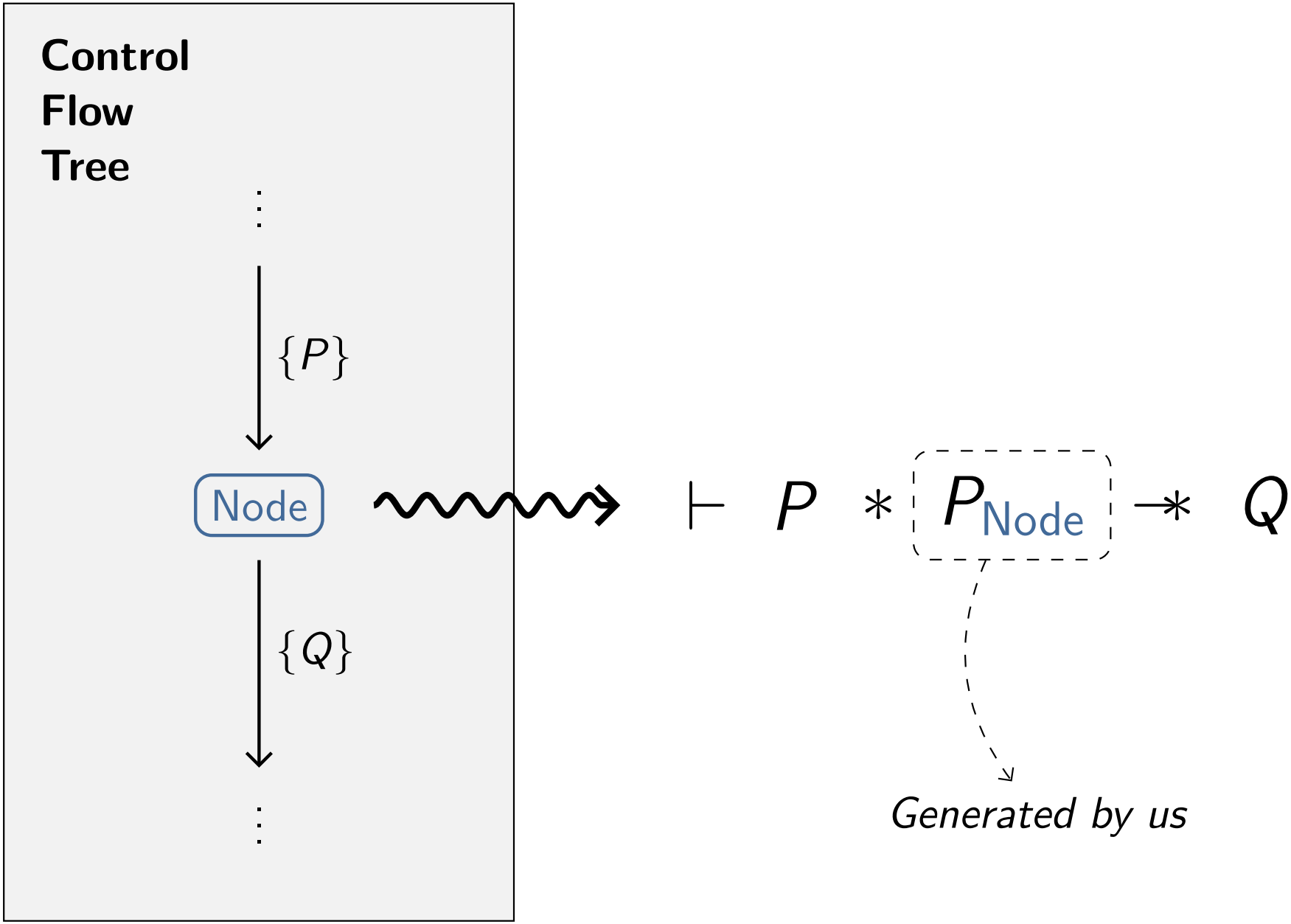


Obligation Generation



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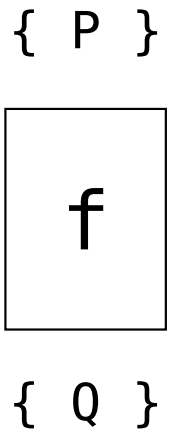
Handling functions

```
let f x =
  { P }
  ...
  { Q }
in
...
```



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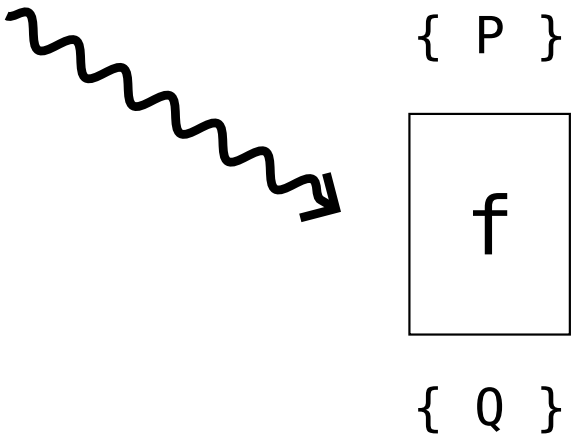

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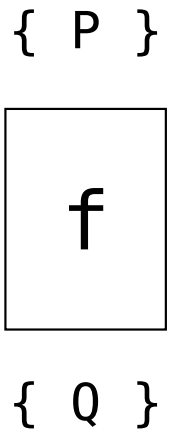

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let f x =
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```
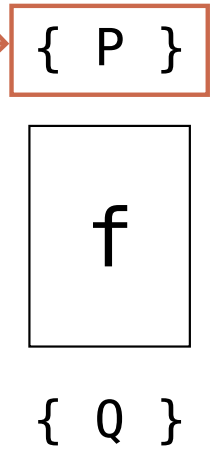

$\{ P[u/x] * R \}$

`f u`



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let f x =
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...
```

$\{ P[u/x] * R \}$

$\boxed{f\ u}$

$\boxed{\{ Q[u/x] * R \}}$

$\{ P \}$

$\boxed{f}$

$\boxed{\{ Q \}}$

Handling functions

```
let f x =
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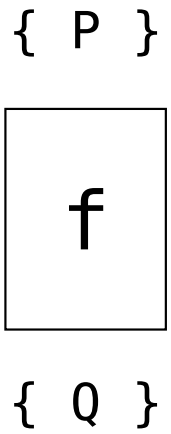
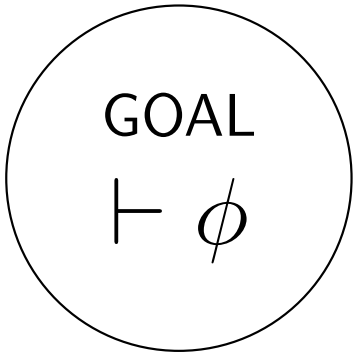


Automatically solving goals: Coq & SMT



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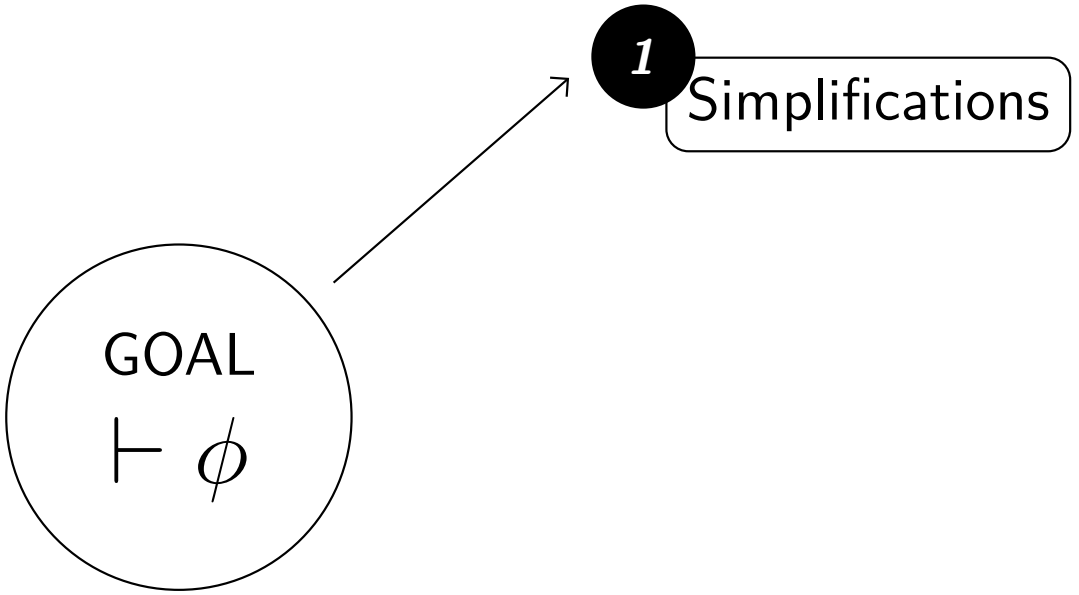
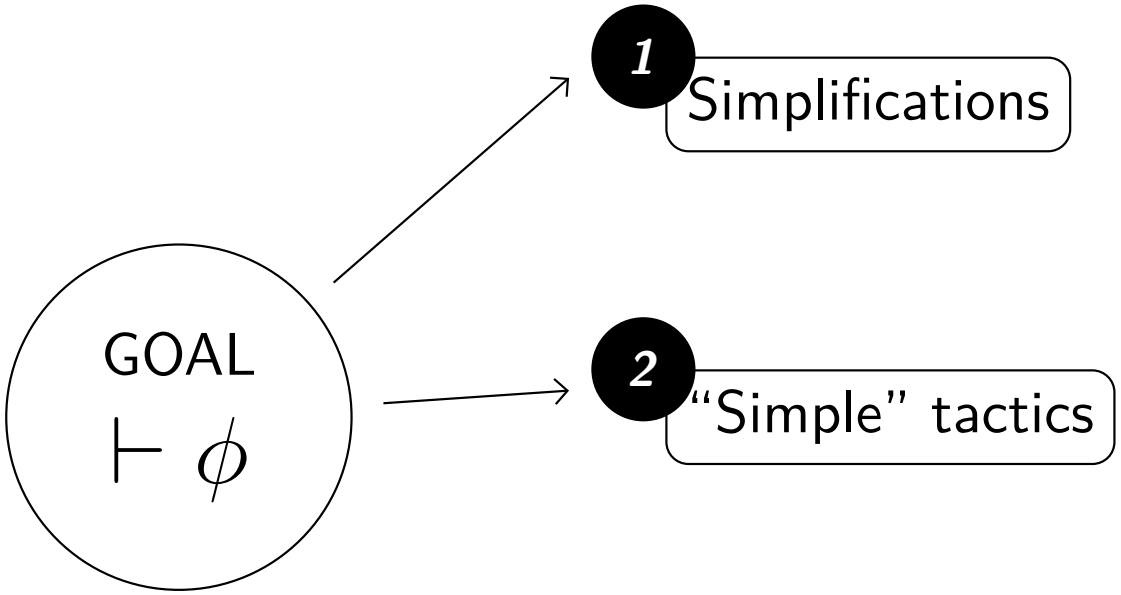


$$\psi * \mathsf{emp} \quad \rightsquigarrow \quad \psi$$

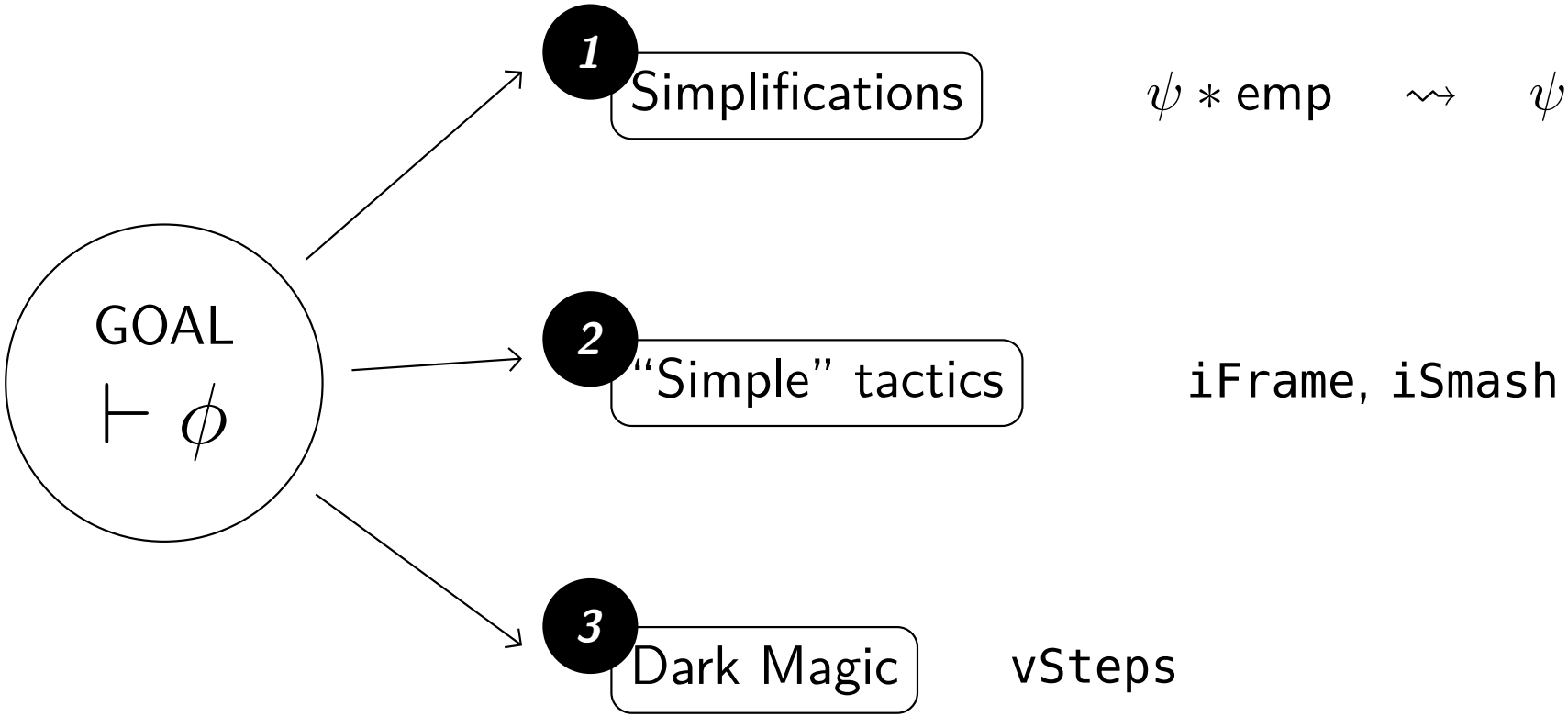
Automatically solving goals: Coq & SMT



$\psi * \mathsf{emp} \quad \rightsquigarrow \quad \psi$

`iFrame`, `iSmash`

Automatically solving goals: Coq & SMT



*preparing the goal for the **SMT solver***

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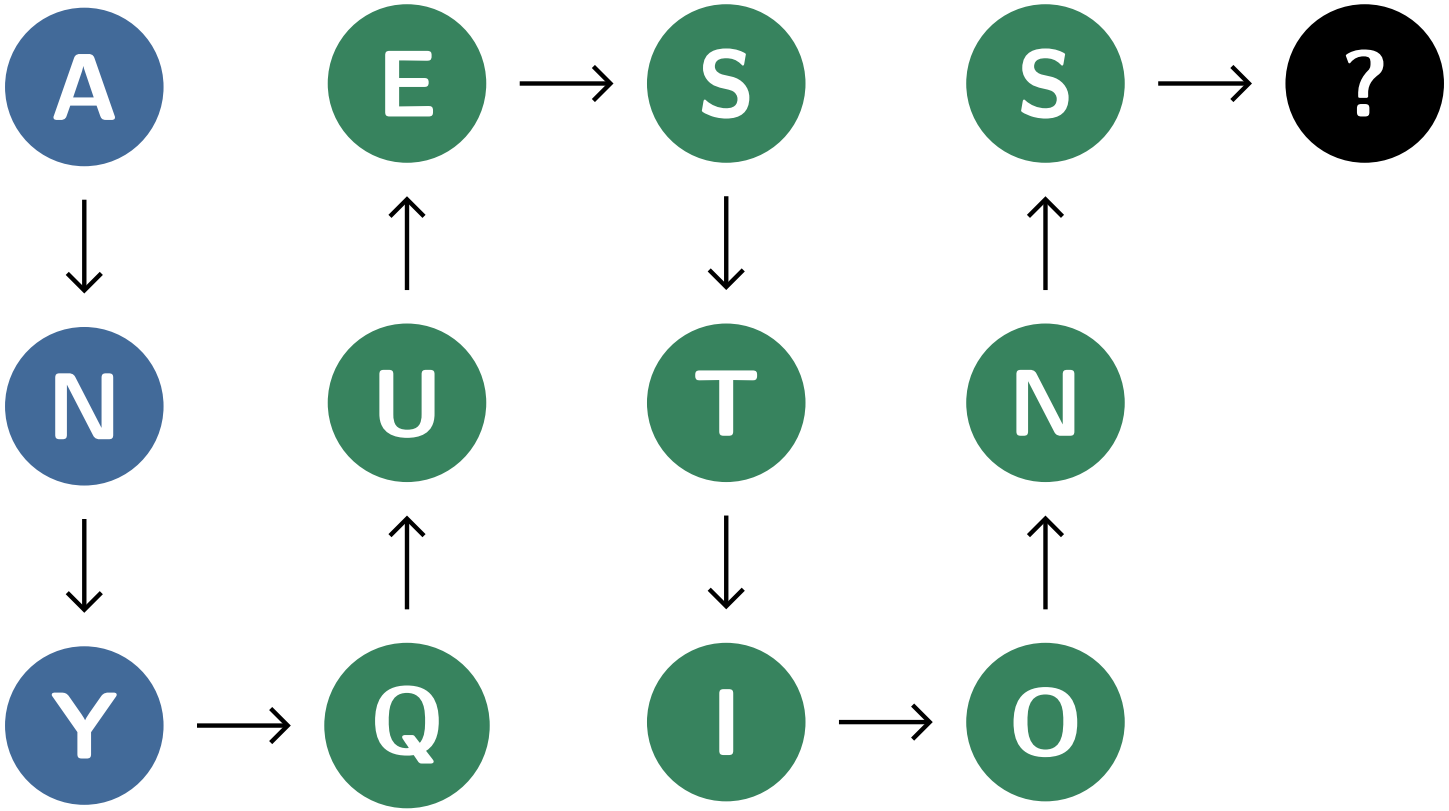
Demo time!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